

Stars and Bars

Activity: Attempt to solve following problems that will explore modifying the stars and bars counting method.

Suppose you have 12 identical balls and 6 numbered boxes.

1. How many ways are there to distribute 12 balls into six numbered boxes if boxes 1 and 2 must be empty?
2. How many ways are there to distribute 12 balls into six numbered boxes if box 1 must contain at least 1 ball and box 2 must contain at least 2 balls?
3. How many ways are there to distribute 12 balls into six numbered boxes if every even numbered box must be empty?
4. How many ways are there to distribute 12 balls into six numbered boxes if every odd numbered box must have at least one ball?
5. How many ways are there to distribute 12 balls into six numbered boxes if exactly two of the boxes must be empty?
6. How many ways are there to distribute 12 balls into six numbered boxes so that at least one is empty?
7. How many ways are there to distribute 12 balls into six numbered boxes so that each box has two balls?